# Adnan Harun Dogan

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## Education

**Master of Science in Computer Engineering** — *Sep 2022 – Jan 2025*
*Middle East Technical University (METU)* — *Ankara, Turkey*
- **Thesis:** Incorporating Piecewise Linear Functions with Constant Regions in Backpropagation
- **Advisor:** Prof. Dr. Sinan Kalkan, Assoc. Prof. Dr. Emre Akbas

**Bachelor of Science in Computer Engineering** — *Sep 2018 – Jul 2022*
*Middle East Technical University (METU)* — *Ankara, Turkey*

## Experience

**Graduate Intern** (Supervisor: Prof. Dr. Christian Holz) — *Mar 2025 – Oct 2025*
*Sensing, Interaction, & Perception Laboratory (SIPLab), ETH Zürich* — *Zürich, Switzerland*
- Developed learned frequency-weighted Kalman filters that denoise the innovation spectrum under spectral supervision
- Built EEG multitaper-spectrum deep models (ConvLSTM) for real-time VR cybersickness detection, +30% over baselines

**Research Assistant** (Supervisor: Prof. Dr. Sinan Kalkan, Assoc. Prof. Dr. Emre Akbas) — *Feb 2023 – Present*
*Image Processing and Pattern Recognition Laboratory (ImageLab), METU* — *Ankara, Turkey*
- Contributed to Bucketed Ranking Loss, speeding up ranking-based training of DETR/transformer detectors by up to 6x
- Developed optimal-transport (Sinkhorn-Knopp) and ranking-based loss formulations for end-to-end object detection

**Undergraduate Intern** (Supervisor: Assoc. Prof. Dr. Erol Sahin, Assoc. Prof. Dr. Bugra Koku) — *Oct 2021 – Feb 2022*
*Center for Artificial Intelligence and Robotics (ROMER), METU* — *Ankara, Turkey*
- Built a self-driving stack for a physical RC car, and trained RL agents for autonomous navigation task in ROS/Gym/PyTorch
- Simulated a Hector-quadrotor swarm in ROS/Gazebo, approximating numerical solvers for multi-agent localization

**Undergraduate Intern** (Supervisor: Prof. Dr. Mohammad Abdullah Al-Faruque) — *Jun 2021 – Sep 2021*
*Cyber-Physical Systems Laboratory, University of California, Irvine* — *Irvine, California, USA*
- Built multi-modal deep models for beat-by-beat cardiac and sleep abnormality detection from physiological signals
- Fused time-series and spectral features for arrhythmia classification, competitive in international challenges

## Publications


**Dogan, A. H.**, Deniz, M., Alemdar, H., & Baran, O. U. (2026). *CoFINN: Conservation flux informed neural networks for physics problems governed by conservation laws.* arXiv'26. arXiv

**Dogan, A. H.**, Demirel, B. U., & Holz, C. (2026). Frequency-weighted neural Kalman filters. ***IEEE/ICRA'26***. arXiv · Code

Demirel, B. U., **Dogan, A. H.**, Rossie, J., Möbus, M., & Holz, C. (2025). Beyond subjectivity: Continuous cybersickness detection using EEG-based multitaper spectrum estimation. ***IEEE/TVCG'25***. DOI · Code

Yavuz, F., Cam, B. C., **Dogan, A. H.**, Oksuz, K., Kalkan, S., & Akbas, E. (2024). Bucketed ranking-based losses for efficient training of object detectors. ***ECCV'24***. DOI · arXiv · PDF · Code

Amin, M. A., **Dogan, A. H.**, Kuru, E. S., Sever, Y., & Angin, P. (2024). Misuse detection and response for orchestrated microservices based software. *AINA'24*. DOI · Data

Karagoz, P., Cekinel, R. F., **Dogan, A. H.**, Oktay, B., Ozturk, A. U., Tonay, T. S., & Tuncel, B. M. (2023). Enhancing underground built heritage analysis with text mining: A case study on Cappadocia. *CNR Edizioni 2023*. ISBN · PDF

Sever, Y., & **Dogan, A. H.** (2023). A Kubernetes dataset for misuse detection. *ITU J-FET 2023*. DOI · PDF · Code

Sever, Y., Ekinci, G., **Dogan, A. H.**, Alparslan, B., Gurbuz, A. S., Jabrayilov, V., & Angin, P. (2022). An empirical analysis of IDS approaches in container security. *IEEE/SRMC 2022*. Best Paper Award DOI

Demirel, B. U., **Dogan, A. H.**, & Al-Faruque, M. A. (2021). Two might do: A beat-by-beat classification of cardiac abnormalities using deep learning and domain-specific features. *CinC 2021*. DOI · PDF · Code

Buyukbas, E. B., **Dogan, A. H.**, Ozturk, A. U., & Karagoz, P. (2021). Explainability in irony detection. *DaWaK 2021*. DOI

**Dogan, A. H.**, & Dogan, A. (2021). *An assembled deep learning approach for flow field prediction.* PDF


## Scores, Honors, & Awards

**University Entrance Exam: ranked 1466th of 2.27M candidates**, OSYM, Turkey — *Aug 2017*

## Technical Skills & Research Interests

**Languages & Tools**: Python/PyTorch, C/C++, ROS2, Rust
**DevOps & HPC**: GitHub, Docker, Slurm (Workload Manager), Apptainer
**Differentiable Optimization**: Combinatorial Optimization, Optimal Transport, Deep State Space Models
**Deep Learning**: Computer Vision, Self-Supervised Learning, Probabilistic Deep Learning, Deep Reinforcement Learning